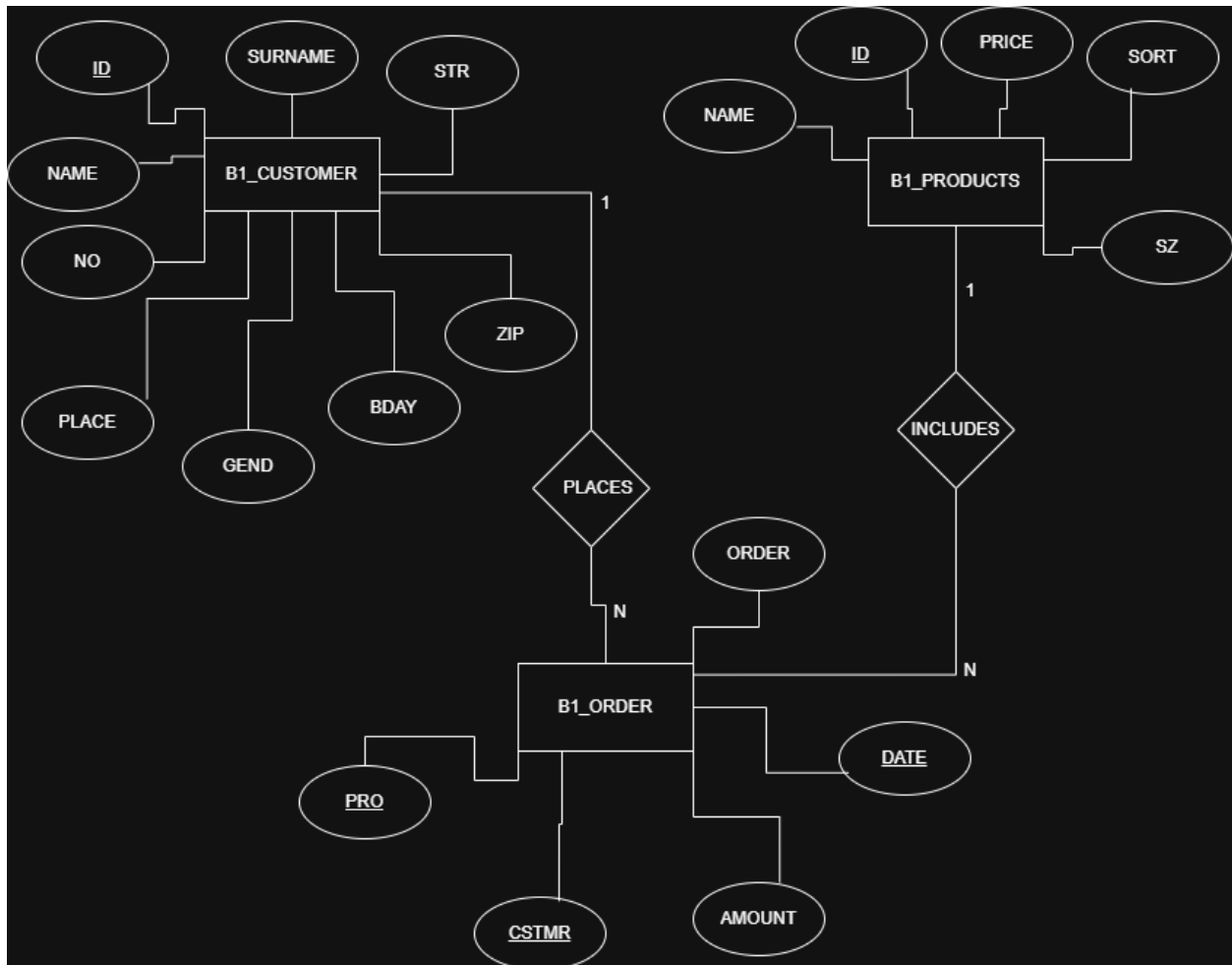


ER DIAGRAM BRANCH 1:

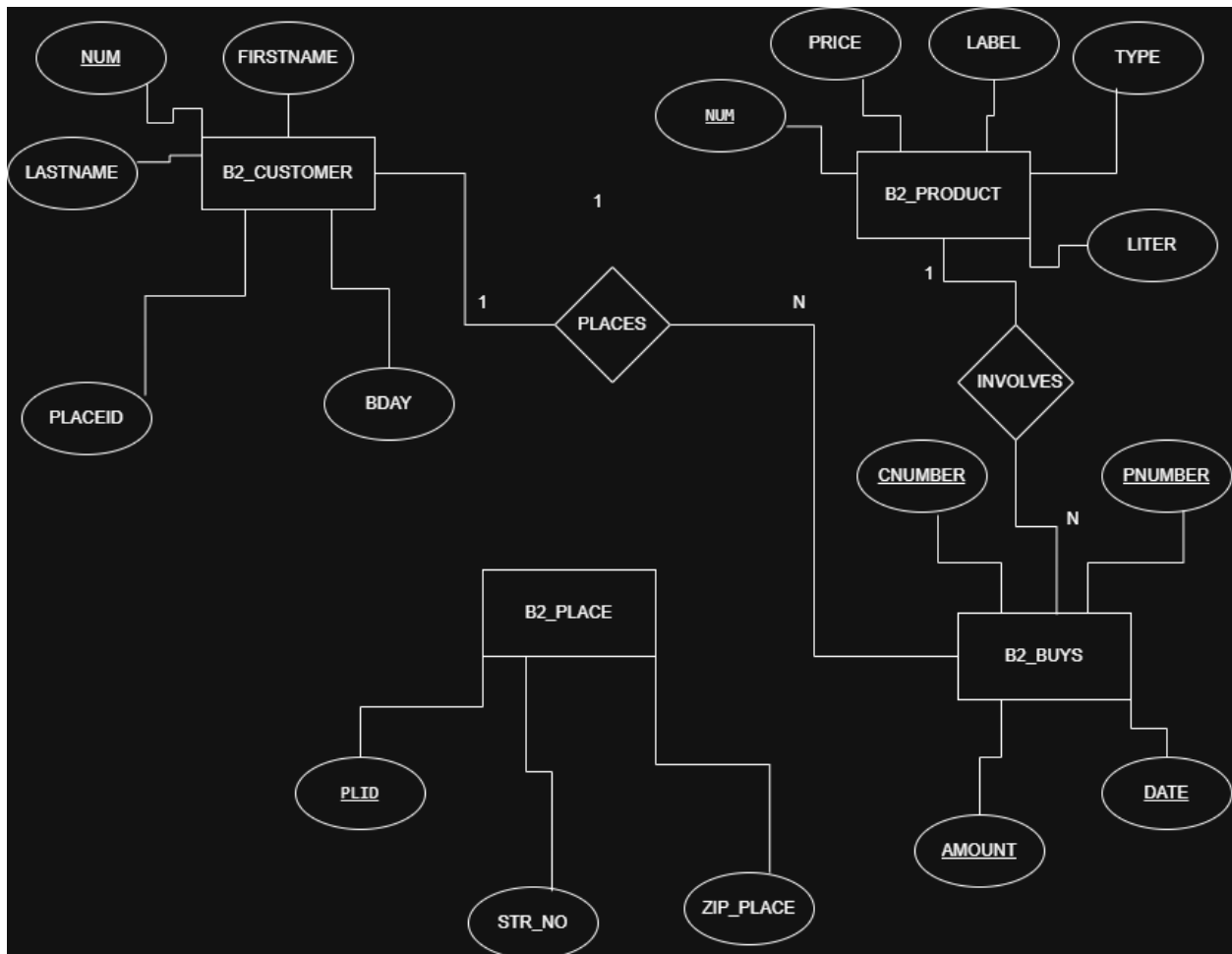


Branch 1: PROBLEM IN DWT_BRANCH_1

- **Lack of Primary Key:** The B1_ORDER table lacks an explicit Primary Key (P). Instead, it relies on a **Composite Unique Constraint (U)** involving DATE, CSTMR, and PRO. This is poor design because it prevents the database from efficiently indexing a single unique ID for transactions.
- **Ambiguous Column Naming:** The use of extremely short column names like PRO (for Product) and CSTMR (for Customer) is inconsistent with standard naming conventions, making the schema harder to read for external developers.
- **Redundant Descriptive Attributes:** Attributes like GEND (Gender) in the CUSTOMER table are stored as separate fields that might be better handled via standardized lookup codes to ensure data integrity.

- ZIP can be a NUMBER instead of VARCHAR2, and if we are using ZIP then we may not need to use PLACE.
- Use of reserved words as column names (ORDER, DATE) in the ORDER table. Also, the table name and column name are the same.

ER DIAGRAM BRANCH 2:



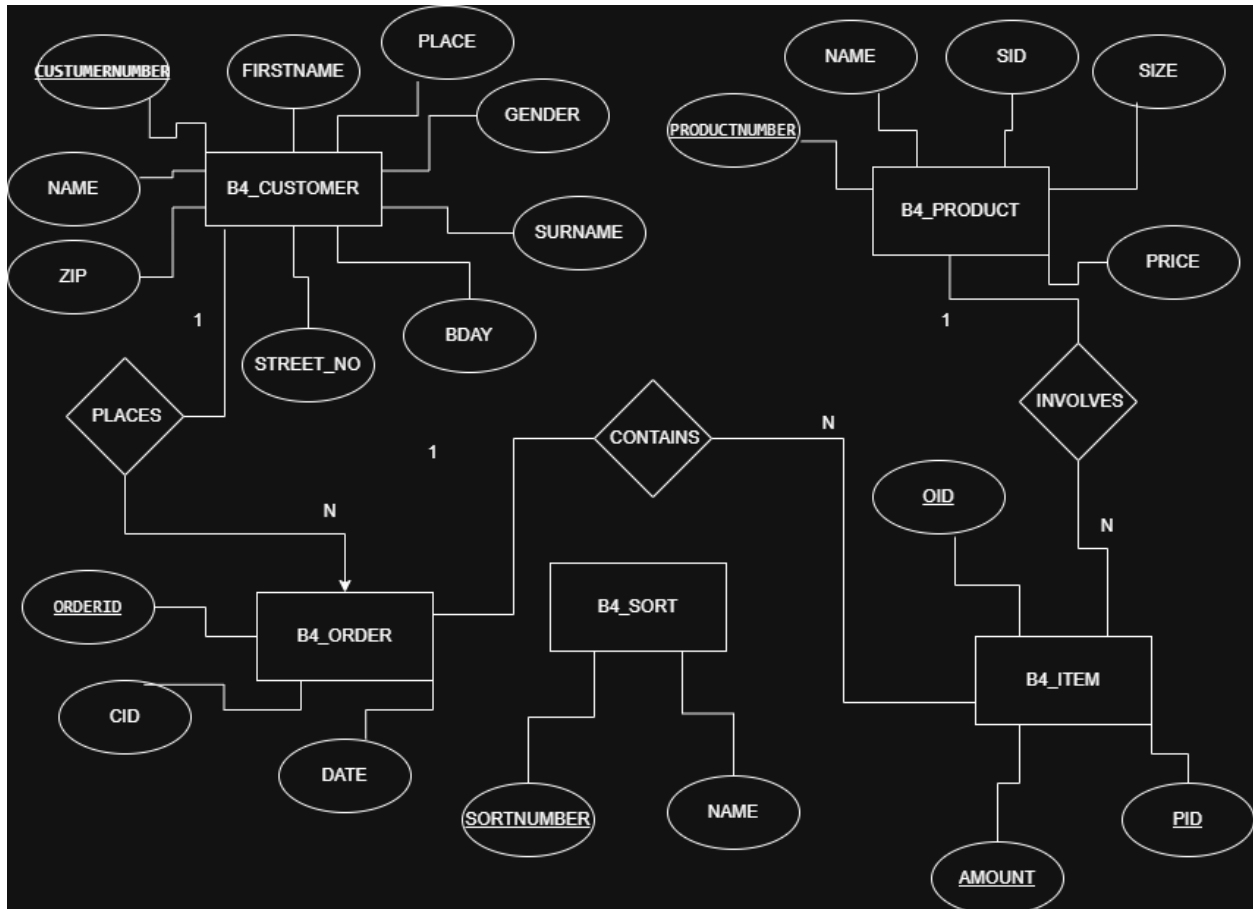
Branch 2: PROBLEM IN DWT_BRANCH_2

- **Over-Complicated Composite Primary Key:** The BUYS table uses a **four-column composite primary key** (CNUMBER, PNUMBER, AMOUNT, DATE). Including AMOUNT in a primary key is a severe design flaw; usually, only the IDs and Date should identify a transaction. If the amount changes, it technically creates a "new" record rather than updating the old one.
- **Inconsistent Data Types:** The AMOUNT column in the BUYS table is defined as **VARCHAR2(5)** (text) instead of a NUMBER type. This is a major error as it prevents

the database from performing mathematical calculations (like SUM or AVG) without manual conversion.

- **Weak Entity Identification:** The CUSTOMER table uses NUM as a primary key. While functional, using a generic term like "NUM" instead of "CUSTOMER_ID" can cause confusion when joining multiple tables that might also have their own "NUM" columns.
- PRICE should be NUMBER instead of VARCHAR2 in the PRODUCT table.
- If FIRSTNAME and LASTNAME are made UNIQUE, then there could be a problem when inserting customers with the same first name and surname into the CUSTOMER table.
- STR_NO is VARCHAR2, and the data is also a string. We can give it a better name instead of suffixing it with "NO" because it creates confusion in the PLACE table.

ER DIAGRAM BRANCH 4:



Branch 4: PROBLEM IN DWT_BRANCH_4

- **Composite Key Overhead:** Similar to the other branches, the ITEM table uses a composite primary key consisting of OID, AMOUNT, and PID. Like Branch 2, including AMOUNT in the primary key is considered poor practice in relational modeling.
- **Flat File Attributes:** In the CUSTOMER table, attributes like PLACE and ZIP are stored directly in the customer's record. In a fully normalized database (3NF), these should be moved to a separate LOCATION table to avoid repeating the city name for every customer in the same zip code.
- **Inconsistent Identifier Names:** The Primary Key for the product table is PRODUCTNUMBER, but in the ITEM table, it is referred to as PID. Using different names for the same data across tables makes the schema more difficult to navigate and maintain.
- Here FIRSTNAME is NOT NULL, while in other branches LASTNAME is NOT NULL. There is inconsistency in the CUSTOMER table.
- Again, STREET_NO with VARCHAR2 creates confusion about whether it expects a number or a string in the CUSTOMER table.
- Use of reserved keyword as a column name (SIZE) in the PRODUCT table.
- Use of reserved keyword "DATE" as a column name in the ORDER table.